

VOLTA – AI Warranty Intelligence Platform

AI-powered warranty claim intelligence and service tracking

CAPSTONE PROJECT

AI + SOFTWARE ENGINEERING



The Problem — and the VOLTA Solution

The Challenge

Customers struggle to understand warranty coverage, track service requests, and manage product claims. The process is fragmented, opaque, and error-prone — leading to frustration and financial loss.

- No unified view of product warranties
- Manual, slow claim processing
- Fraud and abuse go undetected

The VOLTA Solution

VOLTA is an intelligent platform that lets users register products, raise claims, and receive AI-driven warranty insights — all in one place.

- Instant AI-based warranty analysis
- Real-time claim tracking
- Fraud detection and analytics

Key Features & Technology Stack



AI Claim Analysis

Automated warranty eligibility assessment using NLP and rule-based inference.



OCR Invoice Extraction

Extracts purchase details from invoices to auto-populate product records.



QR Product Tracking

QR-linked product profiles enable instant lookup and service history access.



Fraud Detection

Anomaly detection flags suspicious claims before approval.



Analytics Dashboard

Real-time metrics on claims, coverage, and service performance.



Admin Dashboard

Centralized control panel for manufacturers and service teams.



Tech Stack: FastAPI · Streamlit · SQLite · JWT Auth · Docker · AI Integration



Innovation & Impact

VOLTA bridges AI, software engineering, and real-world problem solving — delivering measurable value across the warranty ecosystem.

For Customers

Transparent, instant warranty insights and seamless claim management.

For Manufacturers

Reduced fraud, lower operational costs, and data-driven product feedback.

For Service Teams

Streamlined workflows, automated triage, and full audit trails.